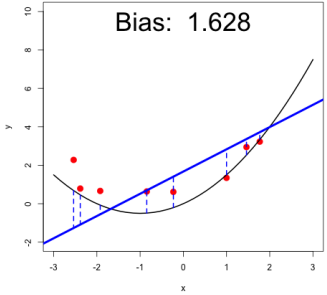
Introduction to Machine Learning

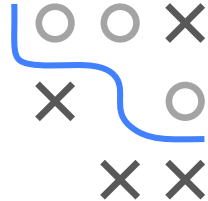
Advanced Risk Minimization
Bias-Variance Decomposition



Learning goals

- Understand how to decompose the generalization error of a learner into
  - bias of the learner
  - variance of the learner
  - inherent noise in the data

BIAS-VARIANCE DECOMPOSITION



Let us take a closer look at the generalization error of a learning algorithm $\mathcal{I}_L$. This is the expected error of an induced model $\hat{f}_{\mathcal{D}_n}$, on training sets of size n, when applied to a fresh, random test observation.

$$GE_n\left(\mathcal{I}_L\right) = \mathbb{E}_{\mathcal{D}_n \sim \mathbb{P}_{xy}^n, (\mathbf{x},y) \sim \mathbb{P}_{xy}}\left(L\left(y, \hat{f}_{\mathcal{D}_n}(\mathbf{x})\right)\right) = \mathbb{E}_{\mathcal{D}_n, xy}\left(L\left(y, \hat{f}_{\mathcal{D}_n}(\mathbf{x})\right)\right)$$

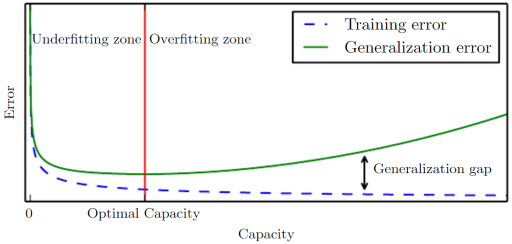
We therefore need to take the expectation over all training sets of size n, as well as the independent test observation.
For the squared loss, there is a nice additive decomposition of $GE_n\left(\mathcal{I}_L\right)$ into three components.*
Hence we assume that the data is generated by

$$y = f_{\text{true}}(\mathbf{x}) + \epsilon\,,$$

with zero-mean homoskedastic error $\epsilon \sim (0, \sigma^2)$ independent of $\mathbf{x}$.

$*$ Similar decomps also exist for other losses expressible as Bregman divergences (e.g. cross-entropy). One exception is the $0/1$ loss (Brown and Ali 2024)

CAPACITY AND OVERFITTING



Credit: Ian Goodfellow

- The performance of a learner depends on its ability to
  1. **fit** the training data well
  2. **generalize** to new data
- Failure of the first point is called **underfitting**
- Failure of the second item is called **overfitting**

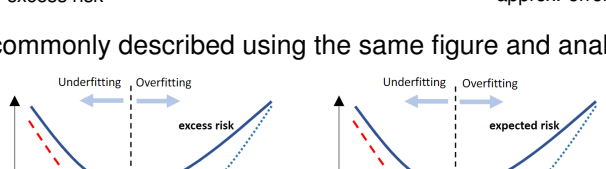
APPROXIMATION AND ESTIMATION

Brown and Ali 2024

The Bias-Variance decomp is often confused or equated with the related (but different) decomp of **excess risk** into **estimation** and **approximation** error.

$$\underbrace{\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f^*_{\mathcal{H}_{all}})}_{\text{excess risk}} = \underbrace{\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f^*_{\mathcal{H}})}_{\text{estimation error}} + \underbrace{\mathcal{R}(f^*_{\mathcal{H}}) - \mathcal{R}(f^*_{\mathcal{H}_{all}})}_{\text{approx. error}}$$

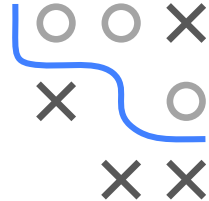
Both are commonly described using the same figure and analogies



Credit: Brown and Ali 2024

NB: It should be noted that the bias-variance decomp. only holds for certain losses, while the above decomposition is universal.

APPROX./ESTIMATION ERROR ▸ Brown and Ali 2024



- The approx. error is a structural property of $\mathcal{H}$.
- The estimation error is random due to dependence on data in $\hat{f}$
- Estimation error arises because we choose $f \in \mathcal{H}$ with limited training data using $\mathcal{R}_{\text{emp}}$ instead of $\mathcal{R}$

Knowing $\hat{f}_{\mathcal{H}} \in \arg\inf_{f \in \mathcal{H}} \mathcal{R}_{\text{emp}}(f)$ assumes we found a global minimizer of $\mathcal{R}_{\text{emp}}$, which is often impossible (e.g. DNNs).

In practice, optimizing $\mathcal{R}_{\text{emp}}$ gives us our 'best guess' $\tilde{f}_{\mathcal{H}} \in \mathcal{H}$ of $\hat{f}_{\mathcal{H}}$. We can now decompose its excess risk finer as

$$\underbrace{\mathcal{R}(\tilde{f}_{\mathcal{H}}) - \mathcal{R}(f^*_{\mathcal{H}_{all}})}_{\text{excess risk}} = \underbrace{\mathcal{R}(\tilde{f}_{\mathcal{H}}) - \mathcal{R}(\hat{f}_{\mathcal{H}})}_{\text{optim. error}} + \underbrace{\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f^*_{\mathcal{H}})}_{\text{estimation error}} + \underbrace{\mathcal{R}(f^*_{\mathcal{H}}) - \mathcal{R}(f^*_{\mathcal{H}_{all}})}_{\text{approx. error}}$$

Note that the optimization error can be negative, even though $\mathcal{R}_{\text{emp}}(\tilde{f}_{\mathcal{H}}) \geq \mathcal{R}_{\text{emp}}(\hat{f}_{\mathcal{H}})$ always holds.